

Experiment No. 8

Inorganic Qualitative Analysis (Salt no. 4)

Aim: Analyse one basic (cation) and one acidic (anion) radical from given inorganic salt.

Apparatus: Test tubes, test tube stand, spatula, glass rod, watch glass, beaker, burner, etc.

A. Preliminary tests: (General scheme)

Test	Observation	Inference
Nature	Crystalline (Soluble in water)	Cl^- , Br^- , I^- , NO_2^- , SO_4^{2-} may be Present
Color of salt	white	Pb^{2+} , Al^{3+} , Mg^{2+} , Ba^{2+} , Zn^{2+} , NH_4^+ , Ca^{2+} , Mn^{2+} , Sr^{2+} Present
Solubility	Soluble in water	crystalline salt Present

B. Dry tests for basic radicals (Cations)

Test	Observations	Inference
1. Heating in a dry test tube:- Take a small quantity of salt in a clean, dry test tube and heat it strongly. Observe the color of the residue and gas evolved.	White infusible residue	Al^{3+} , Zn^{2+} , Ba^{2+} , Ca^{2+} , NH_4^+ may be Present
2. Test for NH_4^+ Salt + NaOH, heat gently, hold moist turmeric paper near the mouth of test tube.	No reaction	-
3. Charcoal cavity test: Take the salt and solid Na_2CO_3 in the ratio 1:2 on watch glass, add few drops of water and prepare a paste, place this in charcoal cavity and heat it with a blow pipe in a reducing flame.	White infusible residue	Zn^{2+} , Al^{3+} , Mg^{2+} , Ba^{2+} , Sr^{2+} , Ca^{2+} may be Present
4. Cobalt Nitrate Test: If white infusible residue is obtained in charcoal cavity test, add two drops of cobalt nitrate solution on the white residue and heat in the blue flame.	Residue turns green.	Zn^{2+} may be Present.

5. Flame test: Clean a platinum wire with conc. HCl. Moisten a small quantity of the substance with concentrated HCl on watch glass. Dip the cleaned platinum wire in it and heat it in oxidizing flame (non - luminous) and observe the color of the flame.		
6. Borax bead test: [For colored salt only] Make a small loop at the end of a clean platinum wire heat it strongly and dip it in borax powder. Heat it again till a glossy bead is obtained. Now touch the hot bead with a small crystal of the salt and heat it again in oxidizing (blue) flame. Observe the color of the bead.		

C. Dry tests for acidic radicals (Anions)

Test	Observations	Inference
i. Action of dil HCl: Salt + dil HCl.	No reaction	CO_3^{2-} , SO_3^{2-} , NO_2^- are absent
ii. Action of conc. H_2SO_4: Salt + conc. H_2SO_4 + heat	No reaction	Cl^- , Br^- , I^- , NO_3^- are absent
iii. Action of MnO_2 and conc. H_2SO_4: Salt + a pinch of MnO_2 + conc. H_2SO_4 and heat	No reaction	Cl^- , Br^- , I^- , NO_3^- , CH_3COO^- are absent
iv. Cu-foil test: (If brown gas is obtained in the above test then only perform this test) Salt + Cu-foil + conc. H_2SO_4 and heat strongly	No reaction	NO_3^- absent

Note: (If the anion is not detected in above dry test, then salt may contain SO_4^{2-} or PO_4^{3-})

D. Preparation of solution (Original Solution): Dissolve about 0.5 gm of salt in 20 mL of water in a 50 mL beaker. This is called original solution (O.S.)

E. Wet test for basic radical (cations)

Detection of groups

Test	Observations	Inference
1. O.S. + dil. HCl	No ppt	Group I absent
2. O.S. + dil. HCl + H ₂ S gas / H ₂ S water solution	No ppt	Group II absent
3. O.S. + NH ₄ Cl + NH ₄ OH till alkaline	No ppt	Group III A absent
4. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + H ₂ S gas / H ₂ S water solution	white ppt	Group III B absent
5. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + saturated solution of (NH ₄) ₂ CO ₃	No ppt	Group IV absent
6. O.S. + NH ₄ Cl + NH ₄ OH till alkaline + Na ₂ HPO ₄	No reaction	Group V absent
7. If all the above groups are absent	—	—

Analysis of Group: After detection of a particular group do analysis of that group.

Analysis of Group III B

Test	Observation	Inference
O.S + NH ₄ Cl + NH ₄ OH + H ₂ S gas/water	white ppt	Zn ²⁺ present
C.T. for Zn²⁺ 1. O.S + dil. HCl + K ₄ Fe(CN) ₆	white ppt	Zn ²⁺ confirmed
2. O.S + Na ₂ HPO ₄ sol ⁿ	white ppt soluble in excess	Zn ²⁺ confirmed

F. Wet test for acidic radicals:

Detection of acidic radical from water solution:

1. Wet tests for CO_3^{2-} , SO_3^{2-} , Cl^- , Br^- , I^-

1. O.S. + AgNO_3	No ppt	CO_3^{2-} , SO_3^{2-} , Cl^- , Br^- , I^- are Present
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Distinction between halides (Cl^- , Br^- , I^-):

Note: (Perform this test only if ppt is obtained for AgNO_3 which is insoluble in dil. HNO_3)

Test	Observation	Inference
O.S. + Chloroform + Cl_2 water in excess (shake well)	No reaction	Halides absent

Note: (If one of the halide radical is present then further analysis is not necessary. Perform confirmatory test for that detected radical.)

2. Wet test for SO_4^{2-} and PO_4^{3-}

(This test is required to perform only, if ppt is NOT obtained in AgNO_3 test)

Test	Observation	Inference
O.S. + $\text{Ba}(\text{NO}_3)_2$	white ppt insoluble in dil. HNO_3	SO_4^{2-} Present

3. WET TEST FOR NO_2^-

Test	Observation	Inference
O.S. + dil. H_2SO_4 + FeSO_4 solution	—	—

4. WET TEST FOR NO_3^-

Test	Observation	Inference
O.S. + 3-4 drops of conc. H_2SO_4 + 2-3 drops of diphenyl amine	—	—

5. WET TEST FOR CH_3COO^-

Test	Observation	Inference
O.S. + FeCl_3 solution	—	—

6. WET TEST FOR $C_2O_4^{2-}$

Test	Observation	Inference
O.S. +dil. Acetic acid + $CaCl_2$		

C.T. for SO_4^{2-}

Test	Observations	Inference
1. 0.5 + Acetic acid + Lead Acetate sol ⁿ	white ppt insoluble in dil. HNO_3 but soluble in hot ammonium acetate	SO_4^{2-} confirmed
2. 0.5 + 2 drops of $KMnO_4$ sol ⁿ + $BaCl_2$ solution	Pink ppt	SO_4^{2-} confirmed

Result:- Given inorganic salt no. 4 contains following

- Basic radical Zn^{2+} (Zinc ion)
- Acidic radical..... SO_4^{2-} (Sulphate ion)
- Molecular formula of the salt..... $ZnSO_4$
- Hence the name of the salt..... Zinc sulphate

Remark and sign of teacher: